**Enrollment No.:** _____

Darshan UNIVERSITY

योगः कर्मसु कौशलम्

**Darshan Institute of Computer Application**

**B.Sc. (IT) | Sem-3 | Summer-2024**

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| **Course Code** | : | 2106CS301 | **Date** | : | 15-05-2024 |
| **Course Name** | : | Computer Network & Network Security | **Duration** | : | 150 Minutes |
| | | | **Total Marks** | : | 70 |

**Instructions:**

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

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|---|---|---|---|
| **Q.1** | **(A)** | Explain Network Layer in detail. | **4** |
| | **(B)** | List all functions of presentation layer and explain any one in brief. | **3** |
| | | **OR** | |
| | | List the functions of Data-link layer and explain any one in brief. | |
| | **(C)** | Write full form of OSI? List seven layers of OSI reference model and explain session layer. | **7** |
| | | **OR** | |
| | | List all functions of transport layer and explain in detail any two in brief. | |
| **Q.2** | **(A)** | Explain multiplexing. List 4 methods of it. | **4** |
| | **(B)** | Write difference between WDM and CDM. | **3** |
| | | **OR** | |
| | | List advantages and disadvantages of TDM. | |
| | **(C)** | Explain circuit switching and packet switching. | **7** |
| | | **OR** | |
| | | Explain FDM. | |
| **Q.3** | **(A)** | Calculate the first address and broadcast address for the network 172.31.0.0/16. | **4** |
| | **(B)** | Draw IPv6 header format. | **3** |
| | | **OR** | |
| | | Draw IPv4 header format. | |
| | **(C)** | A host has been assigned an IP address 192.168.27.5. Determine the class, the number of addresses in the block, the first address, and the last address. | **7** |
| | | **OR** | |
| | | Determine the subnet for the given IP address 216.21.6.0 with 35 hosts in each subnet. | |

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|---|---|---|---|
| **Q.4** | **(A)** | Compare unicast, multicast and broadcast transmission. | **4** |
| | **(B)** | Write advantages and disadvantages of Static Routing. | **3** |
| | | **OR** | |
| | | Define following routing metrics: hop count, delay, latency | |
| | **(C)** | Explain Shortest Path Routing. | **7** |
| | | **OR** | |
| | | Explain Link State Routing. | |
| **Q.5** | **(A)** | Explain the importance of Network Security. | **4** |
| | **(B)** | List the various security methods. | **3** |
| | | **OR** | |
| | | Write benefits of Network Security Policy. | |
| | **(C)** | List Firewall types and explain anyone in detail. | **7** |
| | | **OR** | |
| | | Write short note on VPN. | |

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